English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability Theory and Mathematical Statistics, 2023/2024 Homework 3 *The maximum total across all homework assignments this semester is 50 points, corresponding to a bonus of 0.5 on the tutorial grade. Unless otherwise stated, $m, n, k \in \mathbb{N} = \{1, 2, 3, \ldots\}$; define $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$.*

Problem 1. (2 points) A person starts at $n \in \mathbb{N}$ on the number line, moving one step right with probability $p > 1/2$ and one step left with probability $1 - p$. Let p_n be the probability of reaching 0 when starting at n. Express p_1, p_2, p_3 in terms of p.

Problem 2. (2 points) A rabbit starts at 0 and makes independent jumps uniformly distributed on $[0, 1]$ in the positive direction. The interval $[1 - x, 1]$ is a trap of length $x \in [0, 1]$. What is the probability that the rabbit jumps over the trap?

Problem 3. (6 points; simulating random variables) Assume that we can readily simulate $U(0, 1)$ variables. This may be a mathematical task, constructing deterministic sequences that "look like" independent uniform samples (pseudorandom numbers), or a technological task using physical phenomena.

We study three classical algorithms for simulating a value X from a specified distribution: inversion of F_X, rejection sampling, and the Box–Muller method.

Method 1

1. Let $U \sim U(0, 1)$. What is the distribution of $F_X^{-1}(U)$?[1] Use this to propose simulations of:
   (a) $X \sim \text{Exp}(\lambda)$, $\lambda > 0$;
   (b) $X \sim \text{Cauchy}(c)$, $c > 0$, with $f_X(x) = c/[\pi(c^2 + x^2)]$;
   (c) $X \sim \text{Pareto}(\theta)$, $\theta > 0$, with $f_X(x) = \theta x^{-1-\theta} \mathbb{1}_{\{x \geq 1\}}$.
2. What phenomena do these distributions model, and why? Generate samples and illustrate the law of large numbers and central limit theorem in all three cases.
3. What difficulties arise if this method is used for normal or gamma distributions?

Method 2 The following algorithm, associated with ideas of von Neumann from 1949–1950, simulates a variable with density f_X:

- Choose a variable Y that is easy to simulate, for example by Method 1, with density f_Y satisfying $f_X(x)/f_Y(x) \leq c$ for all x and some $c > 0$.
- Generate a value of Y.
- Independently generate $U \sim U(0, 1)$.
- If $U \leq f_X(Y)/[cf_Y(Y)]$, accept Y as a sample from X; otherwise reject it and repeat.

1. Prove that the algorithm is correct.
2. Generate $X \sim N(\mu, \sigma^2)$. *One possibility is to use an exponential proposal after a suitable transformation.*
3. Let $X \sim \Gamma(\alpha, 1)$, with density $f_X(x) = x^{\alpha-1} e^{-x} \mathbb{1}_{\{x>0\}}/\Gamma(\alpha)$.
   (a) For $\alpha < 1$, consider Y with density
   $$f_Y(x) = \frac{\alpha e}{\alpha + e} \left(x^{\alpha-1} \mathbb{1}_{\{0<x<1\}} + e^{-x} \mathbb{1}_{\{x \geq 1\}} \right).$$
   (b) Show that this is a density. How would you simulate Y using Method 1?
   (c) Can Method 2 be used to generate $\Gamma(\alpha, 1)$ variables for $\alpha < 1$?
   (d) How would you simulate $\Gamma(\alpha, \beta)$ using this method and properties of the gamma distribution?

Method 3

1. Let $X^2 \sim \text{Exp}(1/2)$ and $\Theta \sim U(0, 2\pi)$ be independent. Find the distributions of $Z_1 = R \cos \Theta$ and $Z_2 = R \sin \Theta$. Are they independent?
2. How would you use this to simulate n independent $N(\mu, \sigma)$ variables?

[1] If F_X is not strictly increasing, its ordinary inverse may not exist everywhere. One usually uses the generalized inverse $F_X^{-1}(t) = \inf\{x \in \mathbb{R} : F_X(x) \geq t\}$. For our purposes you may assume an ordinary inverse.

A frequently used distribution (why?) is the lognormal: $X \sim \text{Lognormal}(\mu, \sigma^2)$ when $\ln X \sim N(\mu, \sigma)$. Propose three ways to simulate it.

Problem 4. (2 points) Let X take values in $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$, with $\mathbb{E}[X] = 1$, $\mathbb{E}[X^2] = 2$, $\mathbb{E}[X^3] = 5$. Find the smallest possible value of $\mathbb{P}(X = 0)$.

Problem 5. (3 points) Consider discrete real-valued random variables with finite support, that is, with values bounded above and below by constants. Write $X \leq_c Y$ if $\mathbb{E}[f(X)] \leq \mathbb{E}[f(Y)]$ for every convex $f : \mathbb{R} \to \mathbb{R}$.

1. Prove $f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$ for such X and convex f.
2. Give an example with $X \leq_c Y$ but $X \neq Y$.
3. Prove that $X \leq_c Y$ implies $\mathbb{E}[X] = \mathbb{E}[Y]$ and $D(X) \leq D(Y)$.
4. Prove that $X \leq_c Y$ holds if and only if $\mathbb{E}[X] = \mathbb{E}[Y]$ and, for every $a \in \mathbb{R}$,

$$\int_a^\infty \mathbb{P}(X \geq x)\,\mathrm{d}x \leq \int_a^\infty \mathbb{P}(Y \geq x)\,\mathrm{d}x.$$

Problem 6. (3 points) Let E be finite, $V : E \to 2^E$, and $f : E \to \mathbb{R}$. Call $a \in E$ a local minimum if $f(a) \leq f(b)$ for every $b \in V(a)$. Let $M \geq \sqrt{|E|}$ be an integer and $b_1, \ldots, b_M$ independent and uniform on E. Choose k so $f(b_k) = \min_{1 \leq i \leq M} f(b_i)$. Let $u_0 = b_k$ and, for every $n \geq 0$, set $u_{n+1} = u_n$ if u_n is a local minimum; otherwise choose $u_{n+1} \in V(u_n)$ with $f(u_{n+1}) < f(u_n)$. Prove that u_M is a local minimum with probability at least $1/2$.

Problem 7. (3 points) Let $d, n \in \mathbb{N}$, $n \geq 3$, $G = (\mathbb{Z}/n\mathbb{Z})^d$ and $S = \{\pm e_i : 1 \leq i \leq d\}$, where e_i has coordinate $\overline{1}$ in position i and zeros elsewhere. Let $f : G \to \mathbb{R}$ and X be uniform on G. Prove

$$\mathbb{E}[|f(X) - \mathbb{E}[f(X)]|] \leq \frac{nd}{2} \max_{s \in S} \mathbb{E}[|f(X) - f(X + s)|].$$

Problem 8. (3 points) Let X_n be iid with values in $\mathbb{N}$. Assume $\mathbb{P}(X_1 = 0)\mathbb{P}(X_1 = 1) \neq 0$ and set $S_n = X_1 + \cdots + X_n$. Prove $\mathbb{P}(4 \text{ divides } S_n) \to 1/4$ as $n \to \infty$.

Problem 9. (3 points) Let $Z \sim N(0, 1)$. Prove for $x > 0$ that

$$\frac{\sqrt{x^2 + 4} - x}{2e^{x^2/2}} < \sqrt{2\pi}\mathbb{P}(Z > x) < \frac{1}{xe^{x^2/2}}.$$

Find the first four terms of the asymptotic expansion of $f(x) = e^{x^2/2}\mathbb{P}(Z > x)$ as $x \to \infty$.

Problem 10. (2 points) Let $n > 1$ and let A, B, C be independent and uniform on $\{0, 1\}^n$. For $n > 2$, calculate $p_n = \mathbb{P}(ABC \text{ is an equilateral triangle})$ and determine its asymptotic behaviour.